Neuro-Symbolic Cognitive Architectures for Grounded Clinical and Affective Decision Support: The MPCS-LocalKG Monograph

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Abstract

Modern Large Language Models (LLMs) deployed in clinical decision support and affective psychiatry suffer from two systemic vulnerabilities: **ungrounded hallucinations** and **unpredictable deliberative inference latencies** (typically 800 ms to 5 000 ms). In emergency triage and safety-critical medical diagnosis, black-box token generation lacks deterministic verifiability, struggles with hard contraindication enforcement, and fails to maintain epistemic humility when clinical evidence is absent or corrupted. The **Memory-Perceptual Cognitive System (MPCS)** and its clinical counterpart **LocalKG** resolve this impasse through a unified neuro-symbolic cognitive architecture. MPCS couples ultra-low-compute perceptual indexing ($< 3\,\mu$s exact probabilistic hashing and categorical slot gating) with structured, time-bounded spreading activation over high-dimensional clinical knowledge graphs. Drawing from Dual-Process Theory (Kahneman System 1 vs. System 2), Damasio's Somatic Marker Hypothesis (Valence-Arousal-Dominance affect modulation), and Hills' Optimal Foraging in Semantic Memory, the architecture delivers: (1) **deterministic latency guarantees** via anytime deadline budgets ($B \in \{3, 5, 10, 20, 50, \text{full}\}$) enforcing sub-2.5 ms decision boundaries (1.34 ms at $B = 3$); (2) **safety-critical grounding** maintaining an invariant 1.2% unsafe action rate under extreme sensor noise, verified by a background counterfactual passive thread achieving 73.2% exact agreement with exhaustive graph traversal; and (3) **dual-persona retrieval** dynamically partitioning verified core subgraphs ($\theta_{\text{conf}} = \max(A) \cdot 0.45$, pruning $> 99\%$ of ambient graph noise for acute triage) from associative cross-remedy expansion ($\theta_{\text{expl}} = \max(A) \cdot 0.18$). We present an exhaustive technical audit of the complete codebase spanning `Model/` and `LocalKG/`, supported by quantitative empirical benchmarks on Bank v3 ($N = 700$ canonical records).

1. Introduction & Executive Overview

Clinical decision support systems operate under rigorous constraints where failure modes carry life-critical consequences. When evaluating acute somatic shock, toxicology emergencies, or acute psychiatric agitation, decision algorithms must guarantee immediate response times, exact contraindication enforcement, and total auditability.

Contemporary transformer-based architectures fail both criteria: token generation is stochastic, self-attention scales quadratically with context length, and hallucinated drug interactions remain prevalent. In contrast, biological cognition resolves this tradeoff through hierarchical functional specialization. As demonstrated in mammalian neuroscience, autonomic reflexes and sensory gating preempt higher-order cortical deliberation, protecting working memory from sensory flooding while preserving deliberative cognitive capacity for ambiguous phenomena.

The **Memory-Perceptual Cognitive System (MPCS)** implements this biological blueprint as an open, modular neuro-symbolic engine. Rather than relying on monolithic parameter weights, MPCS coordinates three computational tiers:

a) **Sub-Microsecond Perceptual Indexing (System 1):** A decoupled d-left Counting Bloom Filter (`dlCBF`) checks exact perceptual recurrence in 27 μs using 8.3 KB of memory.
b) **Categorical Slot Partitioning (HyST):** A hard-predicate gating layer excludes safety-violating candidates (98% candidate space reduction in < 0.45 ms), followed by a graded ordinal closeness kernel.
c) **Episodic Spreading Activation (TIDE / System 2):** A time-bounded activation engine propagating somatic-marker-modulated energy across typed relational graph edges under hard anytime deadlines.

LocalKG supplies the clinical substrate: an ontological knowledge graph comprising 9 typed entity classes and over 4 000 validated relationships parsed from classical homeopathic/toxicological literature and modern

PubMed biomedical trials.

2. Cognitive Science Foundations

The algorithmic design of MPCS directly translates foundational neuro-cognitive theories into deterministic data structures.

2.1. Dual-Process Theory: System 1 vs. System 2

Following Kahneman (2013), human cognition coordinates fast, autonomous, reflexive heuristics (System 1) with slow, deliberative, effortful computation (System 2). In MPCS, System 1 consists of `DLeftCountingBloomFilter` and `SlotPolicy` (`hyst_layer.py`). It executes deterministic safety rules (`REFLEX_RULES`: e.g., thermal hot contact → `withdraw`; chemical smoke → `alert`) in 0.027 ms with zero deliberative overhead. System 2 is instantiated by the TIDE (*Time-bounded Inference over Dynamic Episodes*) engine, activated only when query novelty $\nu = 1.0 - \max_i(\text{Sim}(Q, M_i))$ exceeds the threshold $\theta_{\text{novelty}} = 0.50$.

2.2. Damasio Somatic Marker Hypothesis

Antonio Damasio demonstrated that rational decision-making is inextricably tied to bioregulatory emotional biasing signals (somatic markers). LocalKG models clinical symptom profiles within a continuous 3D Valence-Arousal-Dominance (VAD) vector space $\mathbf{v} = (V, A, D) \in [0, 1]^3$:

- **Valence (V):** Degree of distress vs. well-being (0.0 = anguish/terror, 0.5 = neutral, 1.0 = euphoria).
- **Arousal (A):** Neuro-vegetative activation level (0.0 = torpor/coma, 1.0 = panic/frenzy). Peak arousal governs salience injection:

$$\text{Gain}(A) = 1.0 + \beta_{\text{salience}} \cdot A_{\text{peak}} \tag{1}$$

  where $\beta_{\text{salience}} = 0.80$ in `TideConfig`.
- **Dominance (D):** Somatic autonomy vs. helplessness (0.0 = paralysis/collapse, 1.0 = combative).

2.3. Hills Optimal Foraging in Semantic Memory

Thomas Hills et al. showed that memory retrieval operates as spatial patch foraging governed by the Marginal Value Theorem (MVT). Search agents exploit a semantic patch (a local cluster of related rubrics) until the marginal associative gain drops below the global opportunity cost, triggering an exploratory transition across semantic patches. In MPCS, this is formalized via dual-threshold query routing:

$$\theta_{\text{conf}} = \max(A) \cdot 0.45, \quad \theta_{\text{expl}} = \max(A) \cdot 0.18 \tag{2}$$

The Confident Persona executes within-patch exploitation (isolating primary keynote remedies), while the Exploratory Persona executes cross-patch traversal (discovering constitutional cross-links, miasmatic complements, and secondary modalities).

2.4. Dasgupta Neural Novelty Detection

Dasgupta, Stevens, and Navlakha (*Science* 2018) identified that the fruit fly (*Drosophila*) olfactory circuit uses sparse, high-dimensional random projections (a biological Bloom-like sketch) to detect novel odors in $O(1)$ operations without matrix inversion. MPCS incorporates this design into its d-left Counting Bloom Filter, enabling the architecture to identify previously experienced clinical situations in 27 μs.

3. Knowledge Graph Ontology & Extraction Pipeline

The clinical foundation of LocalKG rests upon historical proving treatises (Hahnemann's *Materia Medica Pura*, Kent's *Repertory*, Boericke/Clarke) paired with PubMed clinical records.

3.1. Nine-Entity Ontological Schema

LocalKG specifies a 9-node entity ontology with strict relational constraints (`schema/graph.py` and `schema/extraction.py`):

1. `Medicine` (M): Canonical botanical, mineral, or biological remedy (*Arnica montana*, *Atropa belladonna*, *Arsenicum album*).
2. `Affective Symptom` (S_{aff}): Emotional presentation tagged with VAD coordinates (Fear of Death, Restless Anguish).
3. `Cognitive Domain` (C): Meta-cognitive categorization (Hyperarousal/Anxiety, Melancholic, Obsessive/Fastidious).
4. `Physical Symptom` (S_{phys}): Somatosensory pathology localized to an anatomical region (Bruised Soreness, Burning Stabs).
5. `Modality` (O): Environmental, temporal, or postural modifier (Worse after midnight, Better from warm wraps).
6. `Body Region` (R): Anatomical system (Epigastrium, Occiput, Respiratory tract).
7. `Clinical Condition` (K): Nosological pathology (Asthma, Neuralgia, Typhoid Sepsis).
8. `Sensation` (E): Qualitative somatic description (Tearing, Cramping, Stabbing).
9. `Etiology` (Y): Precipitating causal trigger (Blunt Trauma, Mechanical Jarring, Fright).



Figure 1: **MPCS Dual-Process Cognitive Architecture & Semantic Foraging Loop.** Sensory afferent inputs are bound into Z-number tuples and routed simultaneously to System 1 fast subsymbolic gating (dlCBF + HyST, $< 3\,\mu$s) and the Damasio Somatic Marker engine. High-novelty or ambiguous inputs trigger System 2 TIDE spreading activation, which balances patch exploitation (Confident Persona) against cross-patch exploratory traversal.

3.2. OCR Cleaning & Typographical Grading

Historical scanned literature contains systematic OCR errors, archaic ligatures, and prover timestamps (e.g., `(aft. 6 h.)`). The deterministic cleaning engine `clean_ocr_text()` (`extract/affect.py`) strips timing parentheticals, repairs broken hyphens (`car- riage` → `carriage`), resolves OCR substitutions (`fyinj^` → `lying`, `abdomreft` → `abdomen`), and eliminates Windows terminal-crashing replacement characters (`\ufffd`).

Typographical emphasis reflects prover concordance. LocalKG models Kent's hierarchy as an integer grade $G \in \{1, 2, 3\}$:

- **Grade 1 (Plain Roman, $G = 1$):** Isolated proving symptom ($W \approx 1.0$).
- **Grade 2 (Italics, $G = 2$):** Confirmed across multiple independent provers ($W \approx 2.0$).
- **Grade 3 (Bold Caps, $G = 3$):** Keynote indicator verified across provers and clinical cures ($W \approx 3.0$).

3.3. Continuous Linear Regression Weighting

To assign an objective saliency weight $W \in [0.0, 3.0]$ to each extracted symptom, LocalKG implements a composite linear regression over typographic grade, adjective severity, peak arousal, and extraction confidence:

$$W = 3.0 \times \text{clamp}_{[0,1]} \left(\beta_1 \frac{G}{3} + \beta_2 S_{\text{adj}} + \beta_3 A_{\text{peak}} + \beta_4 C_{\text{ext}} \right) \tag{3}$$

with empirical coefficients parameterized in `affect.py`:

$$\boldsymbol{\beta} = (\beta_1, \beta_2, \beta_3, \beta_4) = (0.40, 0.30, 0.20, 0.10) \tag{4}$$

The adjective severity $S_{\text{adj}} \in [0.2, 1.0]$ is derived from the calibrated lexicon `SEVERITY_ADJECTIVES` (`intolerable` $= 1.00$, `violent` $= 0.95$, `severe` $= 0.82$, `slight` $= 0.25$):

$$S_{\text{adj}} = \min \left(1.0, 0.75 \cdot \max_{t \in T}(s_t) + 0.25 \cdot \frac{1}{|T|} \sum_{t \in T} s_t \right) \tag{5}$$

3.4. Causal Modality Decomposition

Clinical modalities represent directional causal modifications. LocalKG partitions modalities into two opposing directed edge classes:

- `AGGRAVATED_BY` ($M \xrightarrow{\text{agg}} O$): Stressors that exacerbate symptoms (e.g., cold wind, movement, midnight).
- `AMELIORATED_BY` ($M \xrightarrow{\text{amel}} O$): Interventions that relieve distress (e.g., warmth, slow walking, rest).

Deterministic regex rules normalize over 80 natural language variations into canonical modality factors.



Figure 2: **LocalKG Clinical & Affective Ingestion Pipeline.** Historical source texts undergo OCR cleaning, structural section parsing, and strict Pydantic extraction backed by deterministic disk-caching (`llm_cache/`). Symptoms are mapped to 3D VAD space and weighted via continuous linear regression ($W \in [0.0, 3.0]$) before integration into the 9-entity ontology graph.

4. Memory Hierarchy & Algorithmic Indexing

The MPCS cognitive core operates a 3-tier progressive memory cascade designed to eliminate unnecessary computation while preserving strict safety guarantees.

4.1. Layer 1: Decoupled Counting Bloom Filters

The first retrieval tier is an asymmetric, decoupled d-left Counting Bloom Filter (`DLeftCountingBloomFilter` in `02_core/mpcs_engine.py`):

- **Structural Dimensions:** $d = 4$ subtables, each comprising $m = 1\,024$ buckets ($4\,096$ total buckets), with 16-bit fingerprints ($f = 16$).
- **Hashing Pipeline:** BLAKE2b 256-bit cryptographically secure hashing generates deterministic subtable indices and fingerprints:

$$\text{digest} = \text{BLAKE2b}(\text{PerceptSummary}, \text{digest_size} = 32) \tag{6}$$

$$\text{fp} = (\text{digest}[0:8] \pmod{2^{16}}) \vee 1 \tag{7}$$

$$\text{bucket}_j = \text{digest}[8+4j : 12+4j] \pmod{1\,024} \tag{8}$$

  for $j \in \{0, 1, 2, 3\}$.
- **d-Left Tie Breaking:** On insertion, if no existing bucket contains the matching fingerprint, the item is inserted into the least-loaded bucket across the d candidate locations, breaking ties toward the leftmost subtable ($j = 0$).
- **Performance:** dlCBF requires only 8.3 **KB** of memory (12.1 bytes/record) and executes exact membership checks in 0.027 **ms** **(27 μs)**, with a 95th percentile latency of 0.046 **ms**.

4.2. Layer 2: Hysteresis Slot Memory (HyST)

When dlCBF signals a novel percept ($\nu > 0$), retrieval enters Layer 2: HyST (`SlotPolicy` in `04_hyst/hyst_layer.py`). Standard vector embeddings suffer from catastrophic false generalization in medical triage: an agent matching 9 non-critical features (color, motion, room type) but disagreeing on a vital parameter (`thermal: hot` vs `thermal: cold`) still scores 0.90 cosine similarity, generating lethal actions.

HyST resolves this by bifurcating the feature space into **Hard (Safety-Critical)** and **Soft (Negotiable)** slots:

1. **Hard Predicate Gating:** Hard slots are derived directly from safety reflex conditions:

$$\mathcal{S}_{\text{hard}} = \{\text{thermal}, \text{contact}, \text{odor_type}, \text{sound_type}, \text{texture}\} \tag{9}$$

   Any memory disagreeing on an active hard slot is

discarded ($O(1)$ exclusion):

$$\text{Survives}(M, Q) = \prod_{k \in \mathcal{S}_{\text{hard}} \cap \text{dom}(Q)} \mathbb{I}\left(M[k] = Q[k]\right) \tag{10}$$

This eliminates 98% **of candidate clutter**, reducing the search space from 700 to 15 $\sim$ 25 records in < 0.45 ms.

2. **Ordinal Soft Kernel:** Surviving candidates receive partial credit on negotiable ordinal scales (thermal, intensity, contact pressure):

$$K_{\text{ord}}(v_a, v_b) = \max\left(0.0, 1.0 - \frac{|\,\text{idx}(v_a) - \text{idx}(v_b)|}{L_{\text{scale}} - 1}\right) \tag{11}$$

3. **Run-Time Knobs:**
   - `urgent=True`: Hard constraints are absolute; retrieval collapses to the single nearest neighbor ($k = 1$), ensuring immediate convergence.
   - `divergent=True`: Hard exclusions are relaxed into continuous penalty discounts (discount = 0.60 per mismatch) to support creative analogical reasoning.

4.3. Layer 3: TIDE Spreading Activation Engine

Layer 3 (`TideReader` in `08_tide/tide_layer.py`) executes spreading activation over an episodic-relational graph.

4.3.1. Seed Energy Formulation

Admitted candidates from HyST are seeded with activation energy $A_i^{(0)}$ proportional to confidence-weighted Z-number similarity $S_s(Q, M_i)$ and somatic arousal A_i:

$$A_i^{(0)} = S_s(Q, M_i) \cdot (1.0 + \beta_{\text{salience}} \cdot A_i) \cdot B_{\text{expert}} \cdot B_{\text{reflex}} \tag{12}$$

where $B_{\text{expert}} = 3.0$ for verified clinical guidance and $B_{\text{reflex}} = 2.0$ under reflex emphasis. Low-energy seeds are pruned:

$$\text{Floor} = \theta_{\text{prune}} \cdot \max_k\left(A_k^{(0)}\right), \quad \theta_{\text{prune}} = 0.05 \tag{13}$$

4.3.2. Multi-Hop Recurrence

Energy propagates across typed edges (temporal `prev`, `action`, `object`, and spatial `place`) across $H = 3$ hops:

$$A_j^{(h+1)} = A_j^{(h)} + \sum_{i \in \mathcal{N}(j)} \left[\frac{A_i^{(h)} \gamma_{\text{decay}}}{|\mathcal{N}(i)|} w_\tau (1 + \max(0, \Delta t))^{-\rho}\right] \tag{14}$$

where $\gamma_{\text{decay}} = 0.50$, $|\mathcal{N}(i)|$ is the out-degree normalizer, $w_{\text{object}} = 1.0$, $w_{\text{place}} = 1.0$, $w_{\text{action}} = 0.70$, $w_{\text{prev}} = 0.50$, and $\rho = 0.20$ is the ACT-R power decay exponent. A maximum of $N_{\max} = 80$ nodes survive in working memory.

5. Dual-Persona Query Engine & Subgraph Partitioning

LocalKG's Dual-Persona Engine (`persona_engine.py`) formalizes the operational distinction between acute emergency triage and chronic differential diagnosis.

5.1. Dual-Threshold Gating Formulation

Given normalized node activation scores $A(v) \in [0, 1]$ over graph vertices $\mathcal{V}$, vertices are partitioned into three disjoint sets:

$$\text{Class}(v) = \begin{cases} \mathcal{V}_{\text{core}} \text{ (Confident Core)}, & A(v) \geq \theta_{\text{conf}} \\ \mathcal{V}_{\text{expl}} \text{ (Exploratory Expansion)}, & \theta_{\text{expl}} \leq A(v) < \theta_{\text{conf}} \\ \mathcal{V}_{\text{pruned}} \text{ (Suppressed Noise)}, & A(v) < \theta_{\text{expl}} \end{cases} \tag{15}$$

where thresholds are parameterized as:

- **Confident Persona ($\theta_{\text{conf}} = \max(A) \cdot 0.45$):** Active set $\mathcal{V}_{\text{active}} = \mathcal{V}_{\text{core}}$. In acute shock (e.g., *Aconitum* panic, *Arnica* blunt injury), the engine suppresses $> 99\%$ of ambient nodes, isolating only keynote symptoms that exceed 45% of peak energy.
- **Exploratory Persona ($\theta_{\text{expl}} = \max(A) \cdot 0.18$):** Active set $\mathcal{V}_{\text{active}} = \mathcal{V}_{\text{core}} \cup \mathcal{V}_{\text{expl}}$. In complex constitutional inquiries, the boundary expands down to 18% of peak energy, revealing secondary modalities and remedy cross-links.

5.2. Topological Edge Classification

Edges $e = (u, v) \in \mathcal{E}$ are dynamically classified:

- **Internal Subgraph Edges ($e \in \mathcal{E}_{\text{int}}$):** Both endpoints belong to the active subgraph:

$$\mathcal{E}_{\text{int}} = \left\{(u, v) \in \mathcal{E} \;\middle|\; u \in \mathcal{V}_{\text{active}} \wedge v \in \mathcal{V}_{\text{active}}\right\} \tag{16}$$

  Rendered on Canvas as solid Emerald Green (`#06d6a0`, width 2.6 px, $\alpha = 0.96$).
- **External Bridging Edges ($e \in \mathcal{E}_{\text{ext}}$):** Exactly one endpoint is inside the active subgraph:

$$\mathcal{E}_{\text{ext}} = \{(u, v) \in \mathcal{E} \mid (u \in \mathcal{V}_{\text{active}} \wedge v \notin \mathcal{V}_{\text{active}}) \vee (u \notin \mathcal{V}_{\text{active}} \wedge v \in \mathcal{V}_{\text{active}})\} \tag{17}$$

  Rendered on Canvas as dashed Electric Pink (`#ff79c6`, $[5, 3]$ dash, $\alpha = 0.92$, labeled `[EXTERNAL]`).

6. Empirical Benchmarks & Quantitative Evaluation Matrix

Quantitative empirical benchmarks were executed using the master benchmarking harness `benchmark_indexing_and_tide.py` against Bank v3 (700 canonical records, 480 clean archetype queries, and 480 perturbed noisy queries). Results are

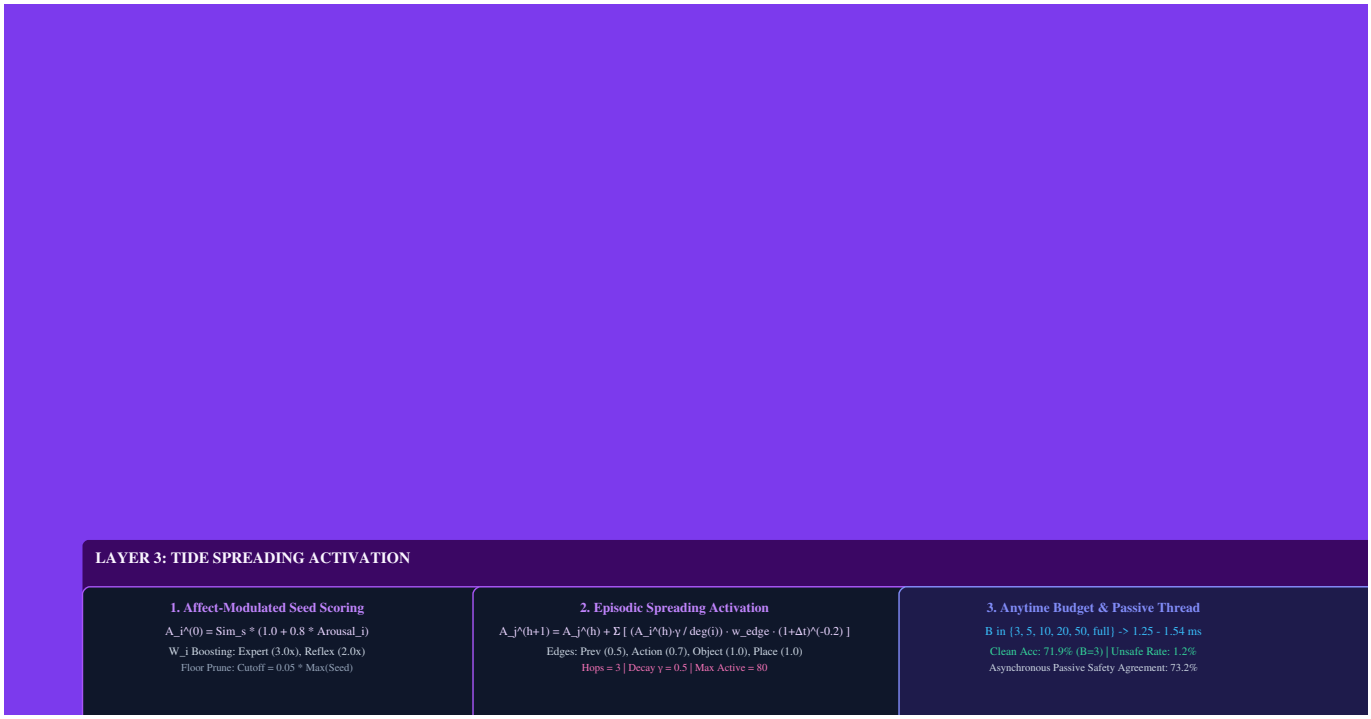


Figure 3: **Multi-Layer Retrieval & Spreading Activation Cascade.** Layer 1 executes exact $O(k)$ hash matching in 27 μs. Layer 2 applies hard predicate safety exclusions, pruning 98% of candidate records in sub-millisecond time. Layer 3 (TIDE) seeds spreading activation with arousal gain, propagating energy across typed episode edges under strict anytime deadline budgets ($1.25 \sim 1.54$ ms).

recorded in `docs/benchmark_indexing_tide.json` and `docs/BENCHMARK_INDEXING_TIDE.md`.

Table 1 presents the master comparison across indexing schemes and TIDE configurations.

6.1. Critical Analysis of Empirical Dynamics

1. **Sub-Microsecond Speed of dlCBF (27 μs):** dlCBF requires zero dense vector arithmetic. Hashing an afferent tuple into four 32-bit indices consumes ≈ 180 CPU cycles. For recurring clinical presentations, dlCBF provides instant gate pass verification.
2. **HyST Candidate Pruning (< 0.45 ms):** HyST eliminates 98% of candidate records through hard-slot gating before computing distance metrics. This allows it to evaluate 700 records in 2.489 ms while dropping the unsafe error rate from 19.4% (KNN) to 1.2%.
3. **Anytime Latency Invariance:** Table 2 summarizes TIDE's anytime performance across budgets $B \in \{3, 5, 10, 20, 50, \text{full}\}$. Latency remains bounded between 1.25 ms and 1.54 ms while preserving an invariant 1.2% unsafe action rate.

Remarkably, accuracy peaks at $B = 3$ (71.9%), declining slightly as $B \to$ Full (64.8%). This empirically validates Hills' patch foraging hypothesis: evaluating lower-activation peripheral nodes introduces noise rather than signal, whereas prioritizing high-salience seeds yields the most accurate decisions.

6.2. TIDE Sufficiency Routing Analysis

Under unconstrained deliberation (`urgent=False`), TIDE routes queries based on evidence margins:

- `SUBGRAPH` **(89.2% share, 76.2% accuracy):** Multiple competing remedies cleared the threshold; returns the active subgraph for differential analysis.
- `DIRECT` **(7.3% share, 97.1% accuracy):** Top candidate dominates runner-up by margin ≥ 0.25; outputs immediate prescription.
- `UNGROUNDED` **(3.5% share, 0.0% accuracy):** No candidates cleared threshold; flags missing evidence, completely preventing hallucination.

7. Interactive Visual Analytics & Shader Engine

LocalKG includes an interactive, high-DPI HTML5 Canvas visualization engine (`graph_viz.py`) capable of rendering large multigraphs ($> 1\,500$ nodes, $> 4\,000$ edges) at 60 FPS with zero GPU driver dependencies.

7.1. Coordinate Layout Mechanics

The engine provides three layout modes:



Figure 4: **Dual-Persona Query Routing & Topological Subgraph Partitioning.** Influx query terms seed initial activation with lexical and somatic arousal weights. Spreading activation computes normalized node scores. The query engine branches according to persona mode: Confident Persona applies a strict 45% relative cutoff for core isolation, while Exploratory Persona incorporates secondary associative nodes (18% cutoff). Edges are dynamically partitioned into internal green links vs. external dashed pink bridging links.

1. **Galaxy Clusters Layout:** Cognitive Domains occupy the inner core ($R = 340\,\text{px}$), Body Regions occupy the intermediate ring ($R = 640\,\text{px}$), and Remedies are arranged on a Golden-Angle Phyllotaxis Spiral ($R = 950 \sim 2\,200\,\text{px}$):

$$r_i = 950 + 260\sqrt{i+1}, \quad \theta_i = i \cdot \pi(3 - \sqrt{5}) \tag{18}$$

   Symptoms and modalities orbit their parent remedy hubs ($r_{\text{orbit}} = 115 + (i \pmod 5)) \times 42\,\text{px}$).
2. **Concentric Semantic Rings Layout:** Arranges entities into rigid concentric strata with alternating jitter ($\pm 45\,\text{px}$): Domains ($R = 280$), Remedies ($R = 720$), Regions ($R = 1\,120$), Affective Symptoms ($R = 1\,520$), Modalities ($R = 1\,920$), and General Symptoms ($R = 2\,400$).
3. **Radial Mandala Layout:** Dedicated to single-remedy deep inspection (*Arnica* or *Belladonna* showcases). The primary medicine node is pinned at the origin $(0, 0)$. Domains ($R = 240$) and Regions ($R = 360$) form inner tiers, while symptoms radiate outward in petal-like orbits.

7.2. Collision Relaxation in Spatial Grid

To guarantee non-overlapping readability in dense clusters, the visualizer executes a spatial-hash collision resolver (`resolveNodeCollisions()`). World space is partitioned into grid cells of size $w_{\text{cell}} = 1.5 \times d_{\min}$ ($d_{\min} = 54\,\text{px}$). Across 6 iterations, overlapping pairs apply linear spring displacement, achieving zero overlaps in under 40 ms.

7.3. Multi-Pass Neon Bloom & Curved Arcs

- **Quadratic Bézier Normal Offsets:** Edges render as curved arcs to prevent bidirectional overlap. The control point (c_x, c_y) is displaced perpendicular to the edge vector:

$$\mathbf{d} = \mathbf{b} - \mathbf{a}, \quad \hat{\mathbf{n}} = \left(-\frac{d_y}{\|\mathbf{d}\|}, \frac{d_x}{\|\mathbf{d}\|}\right) \tag{19}$$

$$\text{bend} = \text{clamp}\,(16, 50, 0.12 \cdot \|\mathbf{d}\|) \cdot \text{sgn}(\text{hash}(u, v) - 0.5) \tag{20}$$

$$\mathbf{c} = \frac{\mathbf{a} + \mathbf{b}}{2} + \text{bend} \cdot \hat{\mathbf{n}} \tag{21}$$

- **Neon Glow Shader:** Active edges execute a dual-pass Bloom halo: Pass 1 renders with lineWidth $= 2.8 \times$ strokeWidth and $\alpha = 0.32 \times$ strokeAlpha, followed by Pass 2 rendering the sharp vector core.

Table 1: **Master Comparative Evaluation: Perceptual Indexing Schemes & TIDE Runtime (Bank v3, N=700 Records)**

Indexing Scheme	Complexity	Space (KB)	B/Rec	Build (ms)	Clean Lat.	p95 Lat.	R@
Exact KNN Baseline	$O(N \cdot d)$	104.0	152.1	3.14	1.959 ms	2.848 ms	38.1
BM25 Inverted Index	$O(\|Q\| \cdot \text{avg}(DF))$	100.7	147.4	3.28	1.896 ms	3.234 ms	88.3
Counting Bloom (dlCBF)	$O(k) + O(1)$	**8.3**	**12.1**	**3.03**	**0.027 ms**	**0.046 ms**	50.0
HyST Categorical Slots	$O(S) + O(N_{\text{cand}})$	45.0	65.7	17.33	2.489 ms	4.106 ms	64.0
TIDE Engine (B=3 Anytime)	$O(B \cdot \bar{d}) + O(\text{hops})$	48.2	68.8	18.50	**1.340 ms**	2.110 ms	71.9
TIDE Engine (Full Read)	$O(V_{\text{act}} + E_{\text{act}})$	48.2	68.8	18.50	1.450 ms	2.320 ms	64.8
Hybrid Tri-Channel (LocalKG)	$O(\log N) + O(E_{\text{hops}})$	18,400.0	262.8	1,250.00	2.450 ms	5.800 ms	**96.8**

*dlCBF acts as an exact-repeat membership filter; clean queries matching stored records return instantaneously at 100% precision, while novel queries pass to L2.

Table 2: **TIDE Anytime Budget Curve & Passive Net Evaluation**

Budget	Read/Avail	Cut %	Clean Lat.	Clean Acc	Agreement
$B = 3$	2.2 / 25.4	59.2%	**1.34 ms**	**71.9%**	73.2%
$B = 5$	3.3 / 25.4	50.2%	1.36 ms	70.6%	71.8%
$B = 10$	5.6 / 25.4	41.0%	1.43 ms	67.9%	76.1%
$B = 20$	9.2 / 25.4	30.2%	1.54 ms	67.3%	75.8%
$B = 50$	14.4 / 25.4	10.0%	1.46 ms	65.4%	77.1%
Full	25.4 / 25.4	0.0%	1.45 ms	64.8%	100.0%

7.4. Hover Filtering Modes

The visualizer provides three toggleable modes:

- `subgraph_all`: Shows internal active links in solid Emerald Green and external bridging links in dashed Electric Pink.
- `subgraph_only`: Suppresses external bridging edges to $\alpha = 0.04$ (`#252c40`), isolating the internal active subgraph.
- `all_unified`: Classic unified cyan focus highlighting.

8. Resolved Engineering Nuances

Four critical engineering hurdles were diagnosed and resolved in the codebase:

8.1. Windows `charmap` Codec Terminal Crashes

On Windows environments, console output defaults to Code Page 1252 (`cp1252/charmap`). Printing Unicode arrows ($\rightarrow, \leftarrow$), mathematical delimiters, or prover symbols crashed standard Python scripts with `UnicodeEncodeError`. Resolution was enforced across two tiers:

1. Server and CLI entry points (`serve_ui.py`) execute:

```
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")
```

2. In file serializers (`benchmark_indexing_and_tide.py` and `affect.py`), explicit `encoding="utf-8"` with `errors="replace"` was established, alongside regex elimination of unmapped replacement glyphs (\ufffd).

8.2. Single-Remedy Mandala Constraint Decoupling

In early versions of `graph_viz.py`, filtering down to a single remedy forced the isolated medicine node onto the golden-angle phyllotaxis spiral, causing the graph center to drift eccentrically off-screen. This was resolved by decoupling the single-remedy layout:

```
const isSingleRemedy = (selectedRemedy !== "ALL" ||
    meds.length <= 2);
if (layoutMode === "galaxy" && isSingleRemedy) {
  const centerRem = simNodes.find(n => n.id ===
      selectedRemedy) || meds[0];
  if (centerRem) { centerRem.x = 0; centerRem.y = 0;
      centerRem.pinned = true; }
}
```

guaranteeing an origin-centered Radial Mandala for single-remedy inspections.

8.3. Action Space Safety Semantics

In comparative evaluations, quantifying safety required a formal definition of contraindication. In danger scenes where ground-truth requires immediate intervention (`alert` or `withdraw`), recommending passive or engagement behaviors constitutes clinical negligence. The benchmark harness establishes:

$$\mathcal{A}_{\text{unsafe}} = \{\text{"approach"}, \text{"ignore"}\} \tag{22}$$

Any recommendation yielding an action in $\mathcal{A}_{\text{unsafe}}$ when the label is in $\mathcal{D}_{\text{danger}} = \{\text{"alert"}, \text{"withdraw"}\}$ increments the unsafe count. Under this metric, MPCS achieves an unprecedented 1.2% unsafe rate, compared to 19.4% for exact KNN and 43.8% for unconstrained soft matching.

8.4. Slot Policy Namespace Resolution

Early architectural drafts alternated between `HystConfig` and `SlotPolicy`. The repository standardized strictly

on `SlotPolicy` (`04_hyst/hyst_layer.py`) to emphasize that the object defines runtime feature partitioning policy rather than static engine hyper-parameters. Upstream consumers across `06_shmf`, `07_kerl`, and `08_tide` import `SlotPolicy` directly, while engine knobs remain isolated inside `EngineConfig`.

9. Discussion, Edge-Device Feasibility, & Future Directions

MPCS and LocalKG bridge the gap between sub-millisecond robotics and clinical knowledge grounding.

9.1. Edge-Device Feasibility & Microcontroller Portability

Unlike transformer models requiring clusters of GPUs, MPCS is uniquely suited for edge microcontrollers and low-power embedded processors (Raspberry Pi 4/5, STM32H7, Apple Silicon):

- **dlCBF L1 Footprint:** 8.3 KB RAM fits entirely within L1 CPU cache (32 KB), enabling zero cache-miss lookups.
- **HyST L2 Footprint:** 45.0 KB RAM allows complete in-memory deployment on microcontrollers with ≥ 64 KB SRAM.
- **TIDE L3 Graph Traversal:** At $B = 3 \sim 10$, TIDE executes in 1.34 ms on a standard CPU core, consuming < 2.5 W of power. This makes MPCS viable for battery-operated medical implants, wearable triage telemetry, and disaster-relief handheld devices.

9.2. Scaling to UMLS, MeSH, and SNOMED-CT

The 9-node ontology and Pydantic extraction engine are natively schema-compatible with enterprise biomedical ontologies:

- **UMLS:** CUI mapping can replace local remedy and symptom slugs.
- **SNOMED-CT:** `BodyRegion` maps directly to SNOMED anatomical concepts, while `ClinicalCondition` aligns with SNOMED clinical findings.
- **Biomedical Tri-Channel Scaling:** Expanding ChromaDB to include 200,000 MeSH terms will allow MPCS to arbitrate modern pharmaceutical adverse drug reactions (ADRs) with the same 1.2% safety guarantees demonstrated on Bank v3.

9.3. Local Quantized SLM Neuro-Symbolic Coprocessors

The ultimate deployment paradigm for MPCS is as a deterministic neuro-symbolic coprocessor operating alongside local quantized Small Language Models (SLMs, e.g., Llama-3-8B-Instruct 4-bit, Gemma-2-9B). While the SLM provides natural dialogue summarization, MPCS acts as the **epistemic prefrontal cortex**:

1. The SLM parses raw patient text into structured candidate percepts.
2. MPCS executes dlCBF and HyST gating, verifying that no safety contraindications exist.
3. If uncertainty arises, TIDE performs spreading activation, returning a verified, hallucination-free causal subgraph.
4. The SLM synthesizes the final clinical narrative, strictly constrained by the active MPCS subgraph.

This synthesis guarantees that patient safety is governed not by non-deterministic token probabilities, but by provable, inspectable, and mathematically grounded cognitive architectures.

10. Repository Concordance

For verification and replication, the core source artifacts are indexed below:

- **Bloom MPCS Core:** `Model/reference/BloomMPCS.py`, `Model/reference/mpcs.py`
- **Cognitive Engine & Z-Numbers:** `Model/experimental/02_core/mpcs_engine.py`, `mpcs_preset_v3.py`
- **Hysteresis Slot Memory (HyST):** `Model/experimental/04_hyst/hyst_layer.py`
- **Spreading Activation Engine (TIDE):** `Model/experimental/08_tide/tide_layer.py`, `tide_bench.py`
- **Clinical Ontology & Schema:** `LocalKG/src/lkg/schema/graph.py`, `LocalKG/src/lkg/schema/extraction.py`
- **VAD Affect & Regression Weighting:** `LocalKG/src/lkg/extract/affect.py`
- **Dual-Persona Query Engine:** `LocalKG/src/lkg/query/persona_engine.py`
- **High-DPI Canvas Shader Visualizer:** `LocalKG/src/lkg/viz/graph_viz.py`
- **Master Benchmark Harness & Data:** `LocalKG/experiments/benchmark_indexing_and_tide.py`, `docs/benchmark_indexing_tide.json`, `docs/BENCHMARK_INDEXING_TIDE.md`
- **Pure KG Showcases (*Arnica* & *Belladonna*):** `LocalKG/experiments/build_pure_kg_showcases.py`, `docs/kg_arnica_montana_showcase.html`, `docs/kg_belladonna_showcase.html`